

GREG DICRISTOFARO

Software Engineer

☎ (215) 510-4508 ✉ gregdicristofaro@gmail.com 🔗 gdicristofaro.com 🔗 github.com/gdicristofaro

SUMMARY

Software Engineer with 8+ years of experience designing, testing, and deploying full-stack applications across a wide range of domains. Expertise spans Java (Spring Boot), Python, C++, TypeScript, C# (ASP.NET Core), modern frontend frameworks (React, Angular), cloud infrastructure (GCP, AWS), and CI/CD practices (Docker, Jenkins, GitHub Actions). Former **educator of 11 years** bringing a track record for collaboration and translating complex technical concepts for non-technical audiences.

EXPERIENCE

SKILLS

Languages: Python, Bash, SQL, C, C++, TypeScript, JavaScript, Java, C#, Go, HTML/CSS

Web/Backend: React, Angular, Node.js, Spring Boot, ASP.NET Core, Server Sent Events (SSE), WebAssembly

Cloud/DevOps: GCP (BigQuery, Cloud Functions, Datastore), AWS (Lambda, DynamoDB, S3, SQS), Azure Blob Storage, Docker, Jenkins, SonarQube, Linux

Databases: PostgreSQL, MS SQL Server, SQLite, MongoDB

Desktop: Tauri, JavaFX, Swing

Development/VCS Tools: Git, GitHub, Claude Code, Copilot, Antigravity, Visual Studio Code, Visual Studio IDE, IntelliJ IDEA IDE

ACHIEVEMENTS

✓ Presenter

Presenter for OSDfCon 2021 (Autopsy Scoring: Finding the Relevant Data with Analysis Results)

✓ KBR Certificate of Appreciation

In recognition of outstanding performance & commitment to excellence (2019)

PROJECTS

constantqjs

Implemented the Constant-Q transform for **spectral analysis** using C++ compiled to **WebAssembly** (via Emscripten), integrating with **TypeScript/Angular** for real-time visualization (Chart.js, Web Audio API)

<https://gdicristofaro.github.io/constantqjs/>

Software Engineer

Sleuth Kit Labs

📅 February 2020 - April 2026 📍 Somerville, MA

- Developed **Digital Forensics and Incident Response software**: Cyber Triage, Autopsy, and The Sleuth Kit in an Agile/Scrum environment
- Expanded and optimized **data pipeline processing** handling complex SQL queries over large data sets using Python, Java, C++
- Created **web crawling and scraping** scripts aggregating open source data for enriching the data pipeline aggregating new features
- Designed and implemented cloud updatable heuristics served to on-premises deployments via **GCP Datastore, BigQuery and Cloud Functions**, then compiled into optimized predicate matching data structures with runtimes in milliseconds for millions of files
- **Modernized testing and CI/CD infrastructure** using Jenkins, Docker, Linux, Bash, GitHub Actions, Python, JUnit/Mockito, SonarQube, and Playwright
- **Standardized multi-platform deployment (Windows, Linux, macOS)** through Snap, Homebrew, and Advanced Installer packages
- Provided internal and **customer documentation** for complex features
- Architected and built **RESTful API and Reactive web applications** using Spring Boot, Swagger, Angular, and React (Material UI), leveraging Server Sent Events (SSE) for real-time data streaming
- Developed parallelized data ingest utilizing **Docker, JMS, ActiveMQ, AWS SQS** for coordination
- Utilized AWS DynamoDB, AWS Lambda creating a **file hash lookup web service** for third-party integration
- Worked with **AWS S3 and Azure Blob Storage** as data sources for upload and download
- Architected **parallel data ingestion system** using Docker, JMS, ActiveMQ, and AWS SQS for coordination, improving performance and scalability of data processing
- Developed third-party integrations in **Go**
- Utilized **Node.js** in web application development, Desktop (Tauri) development, and development utilities
- Developed **Java Desktop Applications** with Swing, JavaFX, Maven, Apache Ivy, Apache POI, SQLite, PostgreSQL, and Netbeans RCP; **Contributed to ControlsFX**, a popular JavaFX controls library

sci-fi-crawl

Sci-fi customizable opening credit text using **React, TypeScript, Material UI, and CSS animations**

<https://gdicristofaro.github.io/sci-fi-crawl/>

PSDtoSVG

Python web application for converting Photoshop files into SVG's for data visualization

<https://gdicristofaro.github.io/PSDtoSVG/>

SeamCarve

Scala implementation of the Seam-Carving Algorithm for content-aware image resizing

<https://github.com/gdicristofaro/SeamCarve>

CSPSolver

Class scheduling program based on the AC-3 algorithm utilizing Apache POI API to import from Excel documents written in Scala

<https://github.com/gdicristofaro/CSPSolver>

CalendarMaker

Calendar making application developed with TypeScript, React, Moment.js, Material-UI, and PptxGenJS

<https://github.com/gdicristofaro/CalendarMaker>

Software Engineer

KBR

📅 July 2017 - February 2020 📍 Lexington Park, MD

- Developed **Single Sign On (SSO) and Role-Based Access Control (RBAC)** web application
- Queried large datasets using Microsoft SQL Server, T-SQL, Entity Framework, and Linq to SQL
- Architected and created tooling to **generate parameterized queries** improving performance from minutes to milliseconds for queries and removed SQL injection threats
- Created **data visualizations** with JavaScript, TypeScript, **D3.js**, and SVG to present complex data to customers
- Led creation and deployment of a **React and ASP.NET Core** web application with CMS and Authentication/Authorization capabilities collaborating with stakeholders and UX designer
- Modernized large J2EE web applications built with JSP and JSF winning an **award of excellence** for customer satisfaction

Music Teacher

St. Mary's County Public Schools

📅 August 2006 - June 2017 📍 Leonardtown, MD

- Built software tools for classroom use, including a **constraint-based scheduling system (CSPSolver)** to automate school scheduling
- Demonstrated strong organizational, leadership, and logistics skills by coordinating school-wide schedules and leading public-facing events
- Created resources and curriculum scaffolded for success at multiple levels

EDUCATION

Master of Science in Software Development

Boston University

📅 September 2017 - May 2019 📍 Boston, MA 🎓 GPA 3.96 / 4.00

- *Classes in web applications using RESTful APIs, Node.js, Express, and MongoDB*
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Software Engineering Certificate

Harvard University

📅 September 2012 - December 2014 📍 Cambridge, MA 🎓 GPA 4.00 / 4.00

Master of Music in Music Education

University of the Arts

📅 June 2012 - December 2015 📍 Philadelphia, PA 🎓 GPA 4.00 / 4.00

- *Designed cross-platform music composition software to foster creativity for young students that processes PCM audio data, visualizes music in a sequencer and renders standard music notation*
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Bachelor of Science in Music Education

Pennsylvania State University

📅 August 2002 - May 2006 📍 University Park, PA 🎓 GPA 3.71 / 4.00